Research Statement

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Ph.D. in Mathematics

1. Research Vision

My research program follows the chain:

$$\Big(\text{FEATURE-CENTRIC THEORY} \xrightarrow{\text{Explain}} \text{AI SYSTEMS}\Big) \xrightarrow{\text{Enable}} \text{RELIABLE CONTROL FROM INTERNALS}$$

where feature-centric theory explains how AI systems learn, and AI systems, once understood, become more reliable from their internals.

The first link is mathematical. Modern AI systems succeed across a remarkable range of domains, yet a principled account of why they succeed remains incomplete. My dissertation introduces a feature-centric framework for the theoretical analysis of neural networks, in place of the conventional weight-centric view. From a single chain-rule identity, it relates gradients in weight space to gradients in feature space, and shows that neural networks trained by gradient descent learn to unpack a nonlinear data manifold, layer by layer, into progressively more linear representations. My proposed research asks what this feature linearization implies for the theory of generalization and for the linear structure of representations.

The second link is more practical. AI systems are increasingly deployed in the real world, yet most alignment methods operate end-to-end: a model is trained to minimize a scalar loss defined on its outputs, and what changes inside it goes unexamined. The resulting alignment is fragile: a single direction in activation space can undo it [2]. My current work turns this feature-centric theory into a transparent, backpropagation-free control algorithm that specifies a desired direction in the hidden states and writes it directly into the weights.

2. Past & Current Research

Past Research

Mathematical Foundations of AI A common thread runs through my mathematical work on AI: the geometry of the data space is the right object of analysis. **ABC3** [5] addresses sample-efficient causal inference in randomized experiments. I recast the imbalance between treatment and control groups as a *distance between their distributions in a kernel-induced feature space*. **Curse of Smoothness** [6] identifies a failure mode in functional neural networks, which enforce smoothness on hidden states and weight functions. I prove that *this functional smoothness bounds the weight gradient norm by a product of Lipschitz constants that compounds across depth*, producing a vanishing-gradient effect that ReLU activations cannot rescue.

Alignment in Practice, and Its Limits Another thread I have been focusing on during my PhD is quantifying and improving the reliability of modern AI systems. I began with a striking observation: pre-trained language models return a **distinctive language modeling distribution for hallucinated texts** [4]. Through extensive analysis, I proposed a simple hallucination mitigation method for generative language models. The **Concordia Contest at NeurIPS 2024** [16] asked participants to build agents that cooperate with other AI agents. Building on the Common Ground concept, learned through sociological training, I won first place out of 197 participants.

Working across theory and practice, I found a significant gap between them: without a proper understanding of how modern AI systems internally process their inputs, we are left with post hoc, output-level benchmarks and no account of why models pass or fail them. Closing this gap requires understanding the mechanism by which deep networks reshape data in the first place: what is the geometry of feature learning?

Current Research: Feature-Centric Framework

In my dissertation and the resulting preprint [7], accepted for NeurIPS 2026, joint work with Adityanarayanan Radhakrishnan (MIT) and Daniel Beaglehole (UCSD), I derived the following exact identity. For any function f of an input $z = Wh$,

$$W^\top \nabla_W f = \nabla_h f \cdot h^\top, \tag{1}$$

which follows from the chain rule alone, with no assumption on architecture, loss, or data. I call this the *feature learning equation*, as it bridges the gradients in weight space and in feature space. Two facts follow from it.

The weight Gram matrix tracks the feature covariance structure. In the dissertation, I show that for a single gradient step $W^+ = W - \gamma \nabla_W \mathscr{L}$, substituting Equation (1) yields

$$(W^+)^\top W^+ - W^\top W = h^+ (h^+)^\top - hh^\top + O(\gamma^2), \qquad h^+ := h - \gamma \nabla_h \mathscr{L}. \tag{2}$$

Here h^+ is the *virtual update*: the hidden state that gradient descent would produce if features were the optimization variable. Although no such update is ever applied, $W^\top W$ tracks the *virtual* covariance structure of the hidden states, and the identity extends to mini-batch training by summing over samples.

Training and layerwise dynamics of feature linearization. Feature linearization in deep networks is known under several names, including linear probing [1], the law of data separation [10], and Neural Collapse [14]. Theoretical understanding beyond idealized settings remains limited, since existing accounts rely on strong assumptions such as the unconstrained features model [13], in which hidden features are directly optimizable.

For hidden states $H \in \mathbb{R}^{d \times N}$ and targets Y, I define *Target Linearity* (TL) as the R^2 of the ridge regressor $\hat{Y}_\lambda = H^\top (HH^\top + \lambda I)^{-1} HY$. TL is defined at every layer and every training step, for any task with a target. I derive the training and layerwise dynamics of this quantity and show that it grows with depth in interpolating networks, so *a deep network sequentially unpacks nonlinear data into progressively more linear representations*. Two known phenomena follow as special cases: Neural Collapse is the extreme of target linearization, and high TL in a latent space is sufficient for linear interpolation in generative models.

3. Proposed Research

Thrust 1. A generalization theory from feature linearization dynamics Classical theory bounds a deep network's complexity globally, and the resulting bounds are loose: the same theory also permits memorization of random labels [18], and no complexity measure predicts generalization reliably [11]. Under the sequential linearization view, each layer has an implicit objective, to return a hidden state from which the target is more linearly recoverable than from the one below, and the work a layer performs is the TL increment it produces. If one layer does nearly all the linearization work, it implements a large transformation and its own generalization error is correspondingly large; if the linearization work is divided, each transformation is small. Depth then helps not only by adding expressivity but also by letting the required nonlinearity be distributed.

> **Conjecture 1** *Generalization is determined by how evenly the work of linearization is distributed across depth. Networks in which every layer contributes a comparable* TL *increment generalize; networks in which the increment concentrates in a few layers memorize.*

This reframing makes classical linear model theory available: bounds for ridge regression are sharp and rarely vacuous. The work proceeds in two steps. First, I will assign each layer a complexity term by treating it as a regression from the features below, measured by the weight norm and feature radius that the surrogate $Y^\top H^\top W^\top W H Y$ already controls, so that a larger TL increment carries a larger term. This turns a trained network into a vector of L numbers rather than a single complexity score. Second, I will define a uniformity metric on that vector and prove that the network's generalization gap is controlled by it, so that a peaked profile gives a worse guarantee than a flat one at equal total work.

Preliminary Results and Supporting Work: In grokking on modular addition, the gap between the middle and last layers narrows sharply at the onset of generalization, as the middle layer takes up a share of the linearization it had not been performing before. The transition is visible in the increment profile, not only in the

final TL value. Independent theoretical support comes from the observation that gradient flow on homogeneous networks keeps the norm gap between adjacent layers bounded throughout training, so layers are automatically balanced [8]. That result concerns the balance of weight norms; whether it extends to a balance of linearization work is what Conjecture 1 asks.

Thrust 2. Linear representation hypothesis from feature linearization Steering, probing, and representation engineering all assume that concepts are encoded as directions in activation space [12, 19]. Yet this assumption holds unevenly: steering vectors are reliable for some concepts and not others [17], some features are intrinsically nonlinear [9], and the layer at which a direction exists is found by trial and error. Because these failures are silent, a practitioner extracting a direction from contrastive examples cannot tell whether it reflects a genuine concept or noise. Feature linearization offers a predictive account. If a network linearizes its input manifolds with respect to a shared target, manifolds carrying distinct concepts are pushed toward a common linear structure, so an affine map between them should exist.

Conjecture 2 *When a network is trained to linearize input manifolds with respect to a target task, there exists an affine transformation mapping the hidden representation of one manifold onto that of the other, where corresponding points are defined by shared task structure (e.g. next token prediction).*

If true, the conjecture predicts which concepts admit linear directions (those sharing task structure) and when the assumption breaks. These predictions determine when the concept directions used in Thrust 3 can be trusted.

Thrust 3. Concept Gradient: Reliable control from model internals Existing alignment methods, from fine-tuning to RLHF, rely on backpropagating a scalar loss or reward. However, a growing body of work reports that this process can optimize superficial proxies of the intended objective (reward hacking [15]), or that training on a narrow domain can shift behavior across unrelated domains (emergent misalignment [3]). These failures expose the limits of output-based training, and suggest that alignment can become more transparent when we can directly specify *where each data point's representation should move*.

Equation (1) suggests a way to do exactly this. For a concept vector s from contrastive texts, setting $\nabla_h f := s$ yields $\Delta W = (W^\top)^\dagger s \cdot h_{\text{cal}}^\top$, where $\dagger$ is the pseudoinverse and h_{cal} are calibration activations. Intuitively, ΔW is the weight update that corresponds to moving the representation h_{cal} in the direction s. Because the update enters the weights, it can be applied repeatedly, controlling behavior directly through the model's internals. I call this backpropagation-free method the *concept gradient*.

Preliminary Results: On Llama3 8B, this update shifts the targeted behavior monotonically over several steps while leaving perplexity and untargeted behavior at baseline, unlike narrow fine-tuning, which spills over into unrelated behavior [3]. More strikingly, when the concept gradient is used to impose the simplex ETF structure of Neural Collapse on the feature space of a randomly initialized network, the resulting training curves closely track those of SGD. This indicates that the concept gradient is not only a practical control tool, but can also serve as a feature-space dual of gradient descent through which neural network training can be analyzed.

4. Long-term Vision

My ultimate goal is to understand how the simplest linear arguments can explain phenomena emerging in complicated modern AI systems. By explaining two prominent phenomena — generalization and linear representations — and constructing a practical control method, I believe the Target Linearity concept can fundamentally change how we think about deep neural networks: though they look inevitably complex, they can be decomposed into much simpler submodules. Because this same linear structure is what makes concepts identifiable and steerable, I see it as the mathematical bridge my research vision promised at the outset: a theory of how networks represent the world is, at the same time, a theory of how to reliably control what they represent.

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